MySQL Introduction (Detailed Guide for PDF)

What is MySQL?

MySQL is an open-source relational database management system (RDBMS) used to store, manage, and retrieve data efficiently.

History of MySQL

- Developed by MySQL AB
- Released in 1995
- Now owned by Oracle Corporation

Why Use MySQL?

- Fast and reliable
- Open source
- Widely used in web applications
- Supports large databases

Features of MySQL

- Relational database system
- High performance
- Secure
- Scalable

Basic SQL Commands in MySQL

Create Database

```
CREATE DATABASE mydb;
```

Use Database

```
USE mydb;
```

Creating a Table

```
CREATE TABLE students (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  age INT
);
```

Inserting Data

```
INSERT INTO students VALUES (1, 'John', 20);
```

Selecting Data

```
SELECT * FROM students;
```

WHERE Clause

```
SELECT * FROM students WHERE age > 18;
```

Updating Data

```
UPDATE students SET age = 21 WHERE id = 1;
```

Deleting Data

```
DELETE FROM students WHERE id = 1;
```

MySQL Data Types

- INT
- VARCHAR
- TEXT
- DATE

- FLOAT

Constraints

- PRIMARY KEY
- FOREIGN KEY
- NOT NULL
- UNIQUE

Joins in MySQL

INNER JOIN

```
SELECT * FROM A INNER JOIN B ON A.id = B.id;
```

LEFT JOIN

```
SELECT * FROM A LEFT JOIN B ON A.id = B.id;
```

RIGHT JOIN

```
SELECT * FROM A RIGHT JOIN B ON A.id = B.id;
```

Aggregate Functions

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

```
SELECT COUNT(*) FROM students;
```

GROUP BY

```
SELECT age, COUNT(*) FROM students GROUP BY age;
```

ORDER BY

```
SELECT * FROM students ORDER BY age DESC;
```

Indexes

```
CREATE INDEX idx_name ON students(name);
```

Views

```
CREATE VIEW student_view AS
SELECT name FROM students;
```

Normalization

Process of organizing data to reduce redundancy and improve efficiency.

Example Project

```
CREATE TABLE employees (
  id INT PRIMARY KEY,
  name VARCHAR(50),
  salary INT
);

INSERT INTO employees VALUES (1, 'Alice', 50000);

SELECT * FROM employees;
```

Advantages of MySQL

- Easy to use
- High performance
- Reliable

Disadvantages of MySQL

- Limited advanced features compared to some DBMS

Best Practices

- Use indexes for performance
- Normalize data
- Backup regularly